

IDENTIFY

Name the OpMode

CONNECT

Map Java names to hardware

WAIT

Pause safely for PLAY

READ

Get fresh joystick values

SET POWER

Send commands to motors

TELEMETRY

Report values to Driver Station

REPEAT

Return to the active check

STOP

Exit the loop and disable

DcMotor

The Java hardware type

leftMotor

Java variable name

rightMotor

Java variable name

hardwareMap

Robot Controller hardware directory

DcMotor.class

Device type to find

"left_motor"

Exact configuration name

"right_motor"

Exact configuration name

Direction.REVERSE

Reverse one side when required

INIT

Begin setup

PLAY

Release the wait gate

STOP

End the OpMode

runOpMode()

Main ordered program method

hardwareMap

Connect configured devices

telemetry.addData()

Prepare a telemetry item

telemetry.update()

Send telemetry to Driver Station

waitForStart()

Wait for PLAY or STOP

CHECK

Is the OpMode active?

READ

Get current joystick values

COMMAND

Set the motor powers

REPORT

Update telemetry

TRUE

Run another cycle

FALSE

Leave the loop

PROGRAM TOKEN

Pass me through each step

JOYSTICK VALUE

Replace me every cycle

RAW VALUE

Value read from the gamepad

MINUS SIGN

Negate: change the sign

POWER VARIABLE

Store the calculated result

-1.0

Fully forward raw Y

-0.7

Forward raw Y example

0.0

Centered stick

+0.4

Backward raw Y example

+1.0

Fully backward raw Y

LEFT POWER

Command for the left motor

RIGHT POWER

Command for the right motor

0.0 / 0.0

Predict both sides stopped

+0.3 / +0.3

Predict gentle forward

-0.3 / -0.3

Predict gentle backward

+0.3 / -0.3

Predict an in-place turn

PREDICT

Say what should happen first

STOP & DISABLE

Before touching or diagnosing

DRIVE

Left-stick Y • forward/back

TURN

Right-stick X • left/right

LEFT POWER $(\text{drive} + \text{turn}) \div \text{denominator}$ **RIGHT POWER** $(\text{drive} - \text{turn}) \div \text{denominator}$ **DRIVE + TURN**

Raw left-side mix

DRIVE - TURN

Raw right-side mix

DENOMINATOR $\max(|\text{drive}| + |\text{turn}|, 1.0)$ **NORMALIZE**

Scale both sides together